

Principal Component Analysis

1 Introduction

Principal Component Analysis (PCA) is a dimensionality reduction technique that projects data onto a lower-dimensional subspace while retaining as much variance as possible. PCA transforms the data into a set of uncorrelated components, called principal components, which capture the maximum variance of the data.

2 PCA Step-by-Step Process

Given a dataset represented by the matrix $X \in \mathbb{R}^{n \times p}$, where n is the number of data points and p is the number of features, the PCA process involves the following steps:

2.1 Step 1: Centering the Data

Subtract the mean of each feature from the dataset. Let μ be the mean vector of X , defined as:

$$\mu = \frac{1}{n} \sum_{i=1}^n X_{i\bullet} \in \mathbb{R}^p \quad (1)$$

The centered data matrix is given by:

$$X_{\text{centered}} = X - \mu \mathbf{1}^\top \quad (2)$$

2.2 Step 2: Covariance Matrix

Compute the covariance matrix S of the centered data. The covariance matrix $S \in \mathbb{R}^{p \times p}$ is defined as:

$$S = \frac{1}{n} X_{\text{centered}}^\top X_{\text{centered}} \quad (3)$$

2.3 Step 3: Eigenvalues and Eigenvectors

Find the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_p$ and their corresponding eigenvectors $v_1, v_2, \dots, v_p$ of the covariance matrix S . These eigenvectors represent the principal components, and the eigenvalues indicate the amount of variance explained by each component.

2.4 Step 4: Projection onto Principal Components (Derivation)

The data is projected onto the first k principal components (where $k < p$) by selecting the top k eigenvectors. The projection of the data onto the principal components is given by:

$$Z = X_{\text{centered}} V_k \quad (4)$$

To derive this, recall that:

- The eigenvectors $v_1, v_2, \dots, v_p$ form an orthonormal basis for the data.
- **When projecting onto the first k principal components, we effectively find each data point's coordinates in this new orthonormal basis.**

Let $A \in \mathbb{R}^{n \times m}$ be a matrix whose columns $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m$ form a basis for some subspace $V \subseteq \mathbb{R}^n$. Given a vector $\mathbf{x} \in \mathbb{R}^n$ that lies in the subspace spanned by the columns of A , we want to find the coordinates $\mathbf{c} = [c_1, c_2, \dots, c_m]^\top \in \mathbb{R}^m$ such that:

$$\mathbf{x} = c_1 \mathbf{a}_1 + c_2 \mathbf{a}_2 + \dots + c_m \mathbf{a}_m = A\mathbf{c}$$

This expresses $\mathbf{x}$ as a linear combination of the columns of A , with coefficients $c_1, c_2, \dots, c_m$.

Finding the Coordinates $\mathbf{c}$

To find the coordinates, solve the system of linear equations:

$$A\mathbf{c} = \mathbf{x}$$

- If A is a **square** and **invertible** matrix (i.e., $m = n$ and A has full rank), then the solution is:

$$\mathbf{c} = A^{-1}\mathbf{x}$$

- If A is **not square** or **not invertible** (e.g., when the columns of A span a subspace of lower dimension), we can solve using the **pseudoinverse** of A , denoted A^+ . The solution is:

$$\mathbf{c} = A^\dagger \mathbf{x}$$

where $A^\dagger = (A^\top A)^{-1} A^\top$, assuming A has full column rank.

Example

Consider $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 2}$, and $\mathbf{x} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix} \in \mathbb{R}^3$. We want to express $\mathbf{x}$ as a linear combination of the columns of A .

The system $A\mathbf{c} = \mathbf{x}$ becomes:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

This results in the following system of equations:

$$c_1 = 3$$

$$c_2 = 2$$

$$c_1 + c_2 = 5$$

From the first two equations, we get $c_1 = 3$ and $c_2 = 2$. The third equation $c_1 + c_2 = 5$ is satisfied. Therefore, the coordinates of $\mathbf{x}$ in the basis spanned by the columns of A are:

$$\mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Thus, $\mathbf{x} = 3\mathbf{a}_1 + 2\mathbf{a}_2$.

Geometric Interpretation

The vector $\mathbf{c}$ gives the coefficients (coordinates) of $\mathbf{x}$ in the basis formed by the columns of A . These coordinates represent how much of each basis vector (the columns of A) is needed to form $\mathbf{x}$.

Projection of a Vector onto the Span of a Matrix

Given a matrix $A \in \mathbb{R}^{n \times m}$, the projection of a vector $\mathbf{x} \in \mathbb{R}^n$ onto the column space (span) of A is given by:

$$\text{Proj}_{\text{span}(A)}(\mathbf{x}) = A(A^\top A)^{-1}A^\top \mathbf{x}$$

This gives the closest vector to $\mathbf{x}$ in the subspace spanned by the columns of A . The vector $\mathbf{c}$, as computed above, gives the coordinates of this projection in the basis formed by A 's columns.

Given that the data is already centered, the projection onto the new subspace is simply the dot product of the centered data matrix with the matrix of eigenvectors. This projects the data into the k -dimensional space spanned by the eigenvectors in V_k .

For a given matrix A , the projection of a vector x onto the span of A is given by:

$$\text{proj}_{\text{span}(A)}(x) = A(A^\top A)^{-1}A^\top x \quad (5)$$

This formula projects x onto the subspace spanned by the columns of A . In PCA, this step is essentially projecting the data points onto the space defined by the principal components. Therefore, the projection of the i th sample point, $X_{i\bullet}^\top$ onto the span of $\{v_1, v_2, \dots, v_k\}$ is given by

$$V_k(V_k^\top V_k)^{-1}V_k^\top X_{i\bullet}^\top = V_k V_k^\top X_{i\bullet}^\top = V_k(X_{i\bullet} V_k)^\top,$$

which shows that $(X_{i\bullet} V_k)^\top$ is the coordinate of the projection of the i th sample $X_{i\bullet}^\top$ onto the span of $\{v_1, v_2, \dots, v_k\}$.

2.5 Example

Consider the following 3×5 matrix X :

$$X = \begin{bmatrix} 2 & 4 & 1 & 3 & 5 \\ 6 & 3 & 2 & 5 & 4 \\ 4 & 5 & 3 & 2 & 6 \end{bmatrix} \quad (6)$$

2.5.1 Step 1: Center the Data

Compute the mean of each column and subtract it from X to get the centered matrix X_{centered} :

$$\mu = [4 \quad 4 \quad 2 \quad 3.33 \quad 5] \quad (7)$$

$$X_{\text{centered}} = \begin{bmatrix} -2 & 0 & -1 & -0.33 & 0 \\ 2 & -1 & 0 & 1.67 & -1 \\ 0 & 1 & 1 & -1.33 & 1 \end{bmatrix} \quad (8)$$

2.6 Step 2: Compute the Covariance Matrix

Now, calculate the covariance matrix S :

$$S = \frac{1}{3} X_{\text{centered}}^T X_{\text{centered}} = \begin{bmatrix} 2.67 & 0.67 & 0.67 & -0.33 & 0.67 \\ 0.67 & 0.67 & 0 & -0.67 & 0.67 \\ 0.67 & 0 & 1 & -0.33 & 0.33 \\ -0.33 & -0.67 & -0.33 & 1.33 & -0.67 \\ 0.67 & 0.67 & 0.33 & -0.67 & 1.67 \end{bmatrix} \quad (9)$$

2.7 Step 3: Eigenvalues and Eigenvectors

The eigenvalues of S are $\lambda_1 = 3.95$, $\lambda_2 = 1.88$, and the corresponding eigenvectors are v_1, v_2 .

2.8 Step 4: Projection

Project the data onto the first two principal components by multiplying the centered data matrix with the eigenvectors corresponding to λ_1 and λ_2 :

$$Z = X_{\text{centered}} V_2$$

The final projected data Z is:

$$Z = \begin{bmatrix} -0.50 & -2.22 \\ 2.17 & -1.53 \\ -1.32 & 3.75 \end{bmatrix}$$

Thus, Z represents the data projected onto a 2D subspace. The result is the projection of the data into a two-dimensional space.

Example

Consider the following matrix $X \in \mathbb{R}^{3 \times 4}$:

$$X = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix}$$

1. Center the data: Compute the mean of each column and subtract it from X . The centered data is:

$$X_{\text{centered}} = \begin{bmatrix} -4 & -4 & -4 & -4 \\ 0 & 0 & 0 & 0 \\ 4 & 4 & 4 & 4 \end{bmatrix}$$

2. Covariance matrix S is calculated as:

$$S = \begin{bmatrix} 16 & 16 & 16 & 16 \\ 16 & 16 & 16 & 16 \\ 16 & 16 & 16 & 16 \\ 16 & 16 & 16 & 16 \end{bmatrix}$$

3. Eigenvalues and Eigenvectors: The eigenvalues are $\lambda_1 = 64$ and $\lambda_2 = 0$, and the eigenvectors are:

$$v_1 = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -0.5 \\ -0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

4. Projection onto the first two principal components:

The final projected data Z is:

$$Z = \begin{bmatrix} -8 & 0 \\ 0 & 0 \\ 8 & 0 \end{bmatrix}$$

Thus, Z represents the data projected onto a 2D subspace.

PCA provides an effective way to reduce the dimensionality of data while preserving as much variance as possible. In this example, we reduced a 3×5 matrix to two dimensions using PCA.

3 Reduced Matrix $\bar{X}$ vs Reconstructed Matrix $\tilde{X}$

In Principal Component Analysis (PCA), two matrices are frequently referenced: $\bar{X}$ and $\tilde{X}$. While they are both derived from the original data matrix X , they serve different purposes in dimensionality reduction and data reconstruction.

- $\bar{X} \in \mathbb{R}^{N \times m}$: The reduced matrix, where m is the reduced number of dimensions (principal components). It represents the data in a lower-dimensional subspace.
- $\tilde{X} \in \mathbb{R}^{N \times p}$: The reconstructed matrix, where p is the original number of dimensions. It represents an approximation of the original data X after projection onto the top m principal components.

This document elaborates on the differences between these matrices and their respective applications.

1. $\bar{X}$: Reduced Matrix

After applying PCA, the original data matrix $X \in \mathbb{R}^{N \times p}$ (with N observations and p features) is projected onto the top m principal components, producing the reduced matrix $\bar{X} \in \mathbb{R}^{N \times m}$, where $m < p$.

$$\bar{X} = XV_m$$

Where:

- $V_m \in \mathbb{R}^{p \times m}$ is the matrix of the top m eigenvectors (principal components).
- Applications of $\bar{X}$:
 1. Dimensionality Reduction:
 - Goal: Reduce the number of features while retaining as much variance as possible.
 - Example: Reducing high-dimensional image data to key components for visualization or clustering.
 - Use: $\bar{X}$ is commonly used for data compression, reducing memory or computational complexity in models.
 2. Feature Extraction:
 - Goal: Extract the most important features from the data.
 - Example: Identifying key patterns in images for object recognition.
 - Use: It is used for applications such as image classification or in text processing to reduce dimensionality.

3. Clustering and Pattern Recognition:

- Goal: Perform clustering on lower-dimensional data to group similar observations.
- Example: Customer segmentation in marketing using $\tilde{X}$ to cluster customers based on key features.
- Use: $\tilde{X}$ helps in simplifying unsupervised learning tasks, making clustering algorithms more effective.

2. $\tilde{X}$: Reconstructed Matrix

After projecting the data onto the top m principal components, the original data can be reconstructed (or approximated) using these components. The reconstructed matrix $\tilde{X} \in \mathbb{R}^{N \times p}$ retains the same dimensions as the original matrix X , but it only captures the information from the top m components:

$$\tilde{X} = XV_mV_m^\top$$

Where:

- $V_m^\top \in \mathbb{R}^{m \times p}$ transforms the reduced data back into the original space.
- $\tilde{X}$ minimizes the Frobenius norm $\|X - \tilde{X}\|_F^2$, ensuring that the difference between the original and reconstructed matrices is minimized.
- Applications of $\tilde{X}$:
 1. Data Denoising:
 - Goal: Remove noise from data while preserving key information.
 - Example: Denoising images by reconstructing the data using only the most significant principal components.
 - Use: $\tilde{X}$ is used in signal and image processing to reconstruct clean versions of noisy data.
 2. Data Reconstruction:
 - Goal: Approximate missing or incomplete data points.
 - Example: Reconstructing missing entries in a dataset based on the patterns learned from the available data.
 - Use: $\tilde{X}$ is used in scenarios where data is incomplete or corrupted, such as medical records or recommendation systems.
 3. Low-Rank Approximation:
 - Goal: Approximate the original data with a lower-rank representation.
 - Example: In recommendation systems, $\tilde{X}$ is used to approximate user-item interaction matrices, capturing only the most important relationships.
 - Use: $\tilde{X}$ is common in lossy compression, such as JPEG, where data is reconstructed with reduced fidelity.

3. Comparison of $\bar{X}$ and $\tilde{X}$

Feature	$\bar{X}$ (Reduced Matrix)	$\tilde{X}$ (Reconstructed Matrix)
Purpose	Dimensionality reduction	Data reconstruction
Size	$N \times m$	$N \times p$
Focus	Reduced dimensions for compression, visualization, clustering	Approximate original data using fewer components
Use Cases	Feature extraction, preprocessing for machine learning, clustering	Denoising, imputation, low-rank approximation
Information Retained	Retains only the most important components (variance)	Approximates the original data with key information retained

Table 1: Comparison of $\bar{X}$ and $\tilde{X}$

Conclusion

Both $\bar{X}$ and $\tilde{X}$ play important roles in the PCA process, each serving different purposes. While $\bar{X}$ is used for dimensionality reduction and feature extraction, $\tilde{X}$ provides a low-rank approximation of the original data, useful for denoising and data reconstruction. Understanding when to use each is crucial in data analysis and machine learning applications.

4 Probabilistic Principal Component Analysis (PPCA)

Probabilistic Principal Component Analysis (PPCA) is a generative model that provides a probabilistic interpretation of Principal Component Analysis (PCA). Unlike traditional PCA, which performs a deterministic projection of the data onto principal components, PPCA adds a noise term and models the data generation process probabilistically. This allows PPCA to deal with noisy data and handle uncertainties better.

The model assumes that each observed data point $x \in \mathbb{R}^D$ is generated from a lower-dimensional latent variable $z \in \mathbb{R}^M$, where $M < D$, via a linear transformation. The goal is to estimate the parameters B (the transformation matrix), μ (the mean), and σ^2 (the noise variance) from the observed data.

5 Probabilistic Model

In PPCA, the relationship between the observed data x and the latent variable z is modeled as:

$$x = Bz + \mu + \epsilon$$

where: - $B \in \mathbb{R}^{D \times M}$ is the matrix that linearly maps the latent variable z to the observed space, - $\mu \in \mathbb{R}^D$ is the mean of the observed data, - $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ is Gaussian noise with variance σ^2 , - $z \sim \mathcal{N}(0, I)$ is the latent variable, which follows a standard normal distribution with mean 0 and covariance I .

The prior distribution of the latent variable z is:

$$p(z) = \mathcal{N}(z \mid 0, I)$$

5.1 Likelihood of Observed Data

The conditional probability of the observed data x given the latent variable z is:

$$p(x \mid z; B, \mu, \sigma^2) = \mathcal{N}(x \mid Bz + \mu, \sigma^2 I)$$

This is a multivariate normal distribution with mean $Bz + \mu$ and covariance $\sigma^2 I$, where I is the identity matrix.

The overall likelihood of the observed data x (marginalizing over the latent variable z) is obtained by integrating over z :

$$p(x \mid B, \mu, \sigma^2) = \int p(x \mid z; B, \mu, \sigma^2) p(z) dz$$

This integral results in a multivariate normal distribution:

$$p(x \mid B, \mu, \sigma^2) = \mathcal{N}(x \mid \mu, BB^\top + \sigma^2 I)$$

where $BB^\top + \sigma^2 I$ is the covariance matrix.

6 Maximum Likelihood Estimation (MLE)

The goal of Maximum Likelihood Estimation (MLE) is to estimate the parameters B , μ , and σ^2 by maximizing the log-likelihood of the observed data $X = \{x_1, x_2, \dots, x_N\}$. The log-likelihood function for the dataset is:

$$\log p(X \mid B, \mu, \sigma^2) = \sum_{i=1}^N \log \mathcal{N}(x_i \mid \mu, BB^\top + \sigma^2 I)$$

This log-likelihood can be written as:

$$\log p(X \mid B, \mu, \sigma^2) = -\frac{N}{2} \log |C| - \frac{1}{2} \sum_{i=1}^N (x_i - \mu)^\top C^{-1} (x_i - \mu) + \text{constant}$$

where $C = BB^\top + \sigma^2 I$. To maximize the likelihood, we compute the MLE estimates for the parameters B , μ , and σ^2 .

6.1 MLE for μ

The MLE estimate for μ is the sample mean of the observed data. By taking the derivative of the log-likelihood with respect to μ and setting it to zero, we get:

$$\mu_{\text{MLE}} = \frac{1}{N} \sum_{i=1}^N x_i$$

Thus, μ is the sample mean of the data.

6.2 MLE for B

The MLE estimate for B is derived by maximizing the log-likelihood with respect to B . After centering the data (subtracting μ from each data point), we compute the covariance matrix of the centered data. The eigen-decomposition of the covariance matrix provides the MLE for B .

Specifically, we compute the covariance matrix S of the centered data:

$$S = \frac{1}{N} X_{\text{centered}}^\top X_{\text{centered}}$$

Then, the MLE estimate for B is given by the top M eigenvectors of the covariance matrix S . Let U_M be the matrix of the top M eigenvectors, and Λ_M be the matrix of corresponding eigenvalues. The estimate for B is:

$$B_{\text{MLE}} = U_M \Lambda_M^{1/2}$$

6.3 MLE for σ^2

The noise variance σ^2 is estimated from the remaining eigenvalues of the covariance matrix. Specifically, it is the average of the eigenvalues not captured by the top M principal components:

$$\sigma_{\text{MLE}}^2 = \frac{1}{D - M} \sum_{j=M+1}^D \lambda_j$$

where λ_j are the eigenvalues of the covariance matrix.

7 Generating New Data Points

Once the parameters B , μ , and σ^2 have been estimated, we can generate new data points from the PPCA model. The process is as follows:

7.1 1. Sampling the Latent Variable z

The latent variable z is sampled from a standard normal distribution:

$$z_{\text{new}} \sim \mathcal{N}(0, I)$$

This is a vector of size M , where each component is sampled independently from a standard normal distribution.

7.2 2. Linear Transformation

Once the latent variable z_{new} is sampled, we compute the corresponding observed data point x_{new} using the linear transformation B and adding the mean μ :

$$x_{\text{new}} = Bz_{\text{new}} + \mu$$

7.3 3. Adding Gaussian Noise

The observed data is noisy, so we add Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ to the generated data point. The final generated data point is:

$$x_{\text{new}} = Bz_{\text{new}} + \mu + \epsilon$$

Thus, the steps for generating a new data point from the PPCA model are:

1. Sample $z_{\text{new}} \sim \mathcal{N}(0, I)$,
2. Compute $x_{\text{new}} = Bz_{\text{new}} + \mu$,
3. Add noise $x_{\text{new}} = Bz_{\text{new}} + \mu + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$.

Probabilistic PCA (PPCA) provides a probabilistic framework for dimensionality reduction that is well-suited for noisy or incomplete data. By introducing a latent variable model with Gaussian noise, PPCA offers a flexible way to handle uncertainty and is a generative model that can be used to simulate new data points. The MLE approach gives exact solutions for the parameters B , μ , and σ^2 , and the model can be used for tasks like data generation, imputation, and denoising.

8 Principal Component Regression (PCR)

Principal Component Regression (PCR) combines Principal Component Analysis (PCA) with linear regression to reduce the dimensionality of data, mitigate multicollinearity, and create a stable model. PCR is especially useful when the number of predictors is large or highly correlated.

Given a data matrix X with n samples and p predictors, and a response vector y , PCR involves:

1. Standardizing X .
2. Performing PCA on X to obtain principal components.
3. Selecting top k components.
4. Using these components to predict y .

8.1 Step 1: Standardization of X

To ensure each feature contributes equally, standardize X :

$$X' = \frac{X - \mu}{\sigma}$$

where μ and σ are the mean and standard deviation of each column in X .

8.2 Step 2: PCA on Standardized Data

Perform PCA on X' by finding the eigenvalues and eigenvectors of the covariance matrix $X'^T X'$:

$$\text{Cov}(X') = X'^T X'$$

Let $V = [v_1, v_2, \dots, v_p]$ be the matrix of eigenvectors. Arrange eigenvalues in descending order, selecting the top k eigenvectors to form V_k .

8.3 Step 3: Scores Matrix (Transformed Data T)

Project X' onto the top k principal components to form the scores matrix T :

$$T = X'V_k$$

8.4 Step 4: Linear Regression on Principal Components

Use T to perform linear regression to predict y :

$$y = T\beta + \epsilon$$

8.5 Step 5: Transformation to Original Predictor Space

Map β back to the original predictor space:

$$\beta_{\text{orig}} = V_k\beta$$

9 Example with 5 Samples and 3 Predictors

Consider a dataset with 5 samples and 3 predictors:

$$X = \begin{bmatrix} 2.5 & 2.4 & 3.5 \\ 0.5 & 0.7 & 1.2 \\ 2.2 & 2.9 & 3.1 \\ 1.9 & 2.2 & 2.8 \\ 3.1 & 3.0 & 3.9 \end{bmatrix}$$

and response vector:

$$y = \begin{bmatrix} 1.0 \\ 0.8 \\ 0.9 \\ 1.1 \\ 1.2 \end{bmatrix}$$

9.1 Step 1: Standardize X

Calculate the mean and standard deviation for each column in X :

$$\mu = [2.04 \quad 2.24 \quad 2.9], \quad \sigma = [0.85 \quad 0.83 \quad 0.83]$$

Standardize X by subtracting μ and dividing by σ :

$$X' = \begin{bmatrix} 0.54 & 0.19 & 0.72 \\ -1.81 & -1.87 & -2.04 \\ 0.18 & 0.79 & 0.24 \\ -0.17 & -0.05 & -0.12 \\ 1.27 & 0.91 & 1.20 \end{bmatrix}$$

9.2 Step 2: Covariance Matrix and Eigen Decomposition

Compute the covariance matrix $X'^\top X'$:

$$\text{Cov}(X') = \begin{bmatrix} 1.0 & 0.8 & 0.9 \\ 0.8 & 1.0 & 0.92 \\ 0.9 & 0.92 & 1.0 \end{bmatrix}$$

Perform eigen decomposition to obtain eigenvalues and eigenvectors. Assume the top 2 eigenvalues are selected with corresponding eigenvectors v_1 and v_2 :

$$V_k = \begin{bmatrix} 0.58 & 0.58 \\ 0.56 & -0.83 \\ 0.58 & 0.58 \end{bmatrix}$$

9.3 Step 3: Project onto Principal Components

Calculate the scores matrix $T = X'V_k$:

$$T = \begin{bmatrix} 0.91 & -0.46 \\ -3.53 & 0.22 \\ 0.53 & 0.08 \\ -0.12 & -0.21 \\ 2.21 & 0.37 \end{bmatrix}$$

9.4 Step 4: Linear Regression on T

Fit a linear regression model with T as predictors for y :

$$\hat{y} = T\beta$$

Compute β using least squares:

$$\beta = (T^\top T)^{-1} T^\top y$$

9.5 Step 5: Transform Coefficients to Original Predictor Space

Map β back to the original predictor space using $\beta_{\text{orig}} = V_k \beta$.

PCR is a useful technique for dimensionality reduction and multicollinearity mitigation. By combining PCA with linear regression, PCR is effective for datasets with numerous predictors or high correlation.